

Robust Tests for Heavy-Tailed Data*

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Abstract

There has been increased interest in economics and finance on the effects of heavy tailed data on the behaviour of test statistics and estimators. First-order large sample theory can be unreliable and lead to inefficient inference when the statistics in question have heavy tails. This paper makes two contributions. First, a lower bound for the rate at which Type-II error converges to zero is provided. While classical results on similar rates of convergence of power focus on situations where exponential rates are feasible, we find the rate to be a much slower polynomial rate when the data have heavy tails. Second, a new procedure based on robust generalized empirical likelihood is developed which leads to test statistics with thinner tails. These tests share the desirable first-order properties that conventional tests, have, while improving their global efficiency properties. Simulation evidence shows that the efficiency gains from the new testing procedure can be substantial. An application demonstrates the ability of the testing procedure to detect assets which tend to carry more or less risk than the market in aggregate.

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1 Introduction

First order asymptotic theory forms the building blocks of modern econometrics. The asymptotic normality of estimators and test statistics is fundamental. Recently, more attention has been paid to higher-order behaviour. In particular, there has been an increased focus on the effects of heavy-tailed data on the finite-sample behaviour of statistics. Recently Müller (2023) showed that the normal approximation to the t-test can be unreliable in small samples when the data have finite variances but infinite third moments. This paper develops a new, but complementary, result on the rate at which Type-II error probabilities converge to 0 when the data are heavy-tailed. We consider hypothesis tests on parameters defined as minimisers of criterion functions. When the influence function of the estimator is unbounded and the data exhibit heavy tails, the rate at which Type-II error converges to 0 is shown to be much slower than the standard exponential rate from the classical theory of large deviations. As an example, we show that the Type-II error of the t-test converges to 0 more slowly than the classic exponential rate when the underlying distributions have heavy tails (see Serfling (2009)). This suggests that while the t-test has nice first-order *local* asymptotic properties the *global* asymptotic properties can be quite poor. Here, we will refer to *local* properties relating to Pitman (1949), Le Cam (2012), and Van der Vaart (2000) sense; see also Newey and McFadden (1994), Lehmann and Romano (2005), and Serfling (2009). By *global* properties, we mean large-deviation.

With these *global* properties in mind, we show that it is feasible to construct a test statistic using Generalised Empirical Likelihood (GEL) which has the same distribution as standard test statistics in applied research under local alternatives, and achieves a faster rate of convergence for Type-II errors under fixed alternatives. To be precise, the local power function of the new test statistic is identical to the standard asymptotic power curves when standard Wald and Rao score tests are considered. Furthermore, when the data have heavy tails, under a fixed alternative the Type-II error rate of the new test statistic $\hat{\eta}$ when compared with the corresponding error rate of a Wald or Score test satisfies

$$\frac{\log P(n\hat{\eta} < C_\alpha)}{\log P(nT_n < C_\alpha)} \rightarrow \infty.$$

This implies that as power approaches 1 as sample increases, there is a sample size n^* such that the power of the test using $\hat{\eta}$ exceeds that from using T_n for all $n > n^*$.

The first main contribution of this paper is establishing a rate function $\Lambda(n)$ such that

$$\lim_{n \rightarrow \infty} -\frac{\log P(nT_n < C_\alpha)}{\Lambda(n)} \leq c$$

for common test statistics T_n . This provides an upper-bound on the performance we can expect from such tests under a fixed alternative. We do this by deriving a lower bound for the probability $P(nT_n < C_\alpha)$. This can be achieved using techniques from extreme value theory. The intuition is that when the data have heavy tails (which we consider to be the case when the moment generating function does not exist) then the probability of a rare event occurring is driven by the extreme order statistics from the sample. This contrasts with the thin-tailed case in which the moment-generating function of the data is finite. This intuition matches well the experience of practitioners in which they are concerned about the robustness of results to a few extreme observations.

In this paper, we say that a random variable X is heavy-tailed when for any $\lambda > 0$,

$$\lim_{x \rightarrow \infty} \frac{-\log(P(|X| > \lambda x))}{n} = 0.$$

One implication of this fact is that the moment generating function of X does not exist, and in fact we have that for any $\epsilon > 0$,

$$\sup_{t \in (-\epsilon, \epsilon)} \mathbb{E} e^{Xt} = \infty.$$

Our definition of heavy tails includes many common distributions, including the Pareto, student-t, and log-normal families. For all of these distributions, the moment generating function does not exist. When the moment generating function exists, classic large deviations state that the rate function $\Lambda(n)$ can be chosen to be n , and thus it is said that the probability in question disappears at an exponential rate in sample size; see, for example [Dembo and Zeitouni \(2009\)](#). We show that in our class of heavy-tailed distributions, under a general set of regularity conditions this rate is much slower, with the worst-case rate in the class we consider being a polynomial rate in sample size, or equivalently $\Lambda(n) = \log n$.

Our results complement existing results in the statistical and econometric literature on approaches to working with heavy-tailed data. Most of this has focused on first-order properties of test statistics when the distribution of the observations exhibit heavy tails, and especially properties of test statistics under the null hypothesis. A lot of work has focused specifically on the t-statistic and other studentized statistics. Under symmetry conditions [Efron \(1969\)](#) discusses that the t-test will tend to be asymptotically conservative when the tails of the data are sufficiently heavy. In [Giné et al. \(1997\)](#), necessary and sufficient conditions for the t-test statistic to be asymptotically standard-normal or subgaussian are provided. In [Shao \(1999\)](#) large-deviation results for the t-test statistic are established. [Bentkus et al. \(2007\)](#) establishes the limit distribution of the t-statistic under fixed alternatives. More

recently, Müller (2019) and Müller (2023) establish that errors in the normal approximation to the distribution of the t-statistic can be slow to disappear under the null hypothesis when the data exhibit Pareto-like tails. We find that similar slow-convergence results hold when looking at the Type-II error rate under a fixed alternative. In a simulation study, Young (2022) argues that not only are heavy-tailed data prevalent in economic applications, but heteroskedastic-robust inference is particularly sensitive to heavy-tailed data, and discusses the implications for power. To our knowledge, this is the first paper in this literature to focus on the theoretical consequences for power of test statistics when the data exhibit heavy tails under the conventional Neyman-Pearson testing regime (i.e. with fixed size).

The second key contribution of this paper is the construction of a robust test statistic which avoids the slow convergence rates. The new procedure preserves the local power properties of standard asymptotically efficient tests. In addition, the new procedure provides for faster rates of convergence for Type-II error to zero under fixed alternatives. Our strategy is based on the bounded tilting GEL procedure in Schennach and Wahlstrom (2024). That paper focused on constructing procedures which are robust to misspecification, while still having higher-order efficiency properties relative to other GEL procedures. We take a different perspective, and choose among the class of bounded-tilting procedures such that the resulting test statistic we construct has the same first-order properties as standard test statistics, while having superior properties from a large-deviations perspective.

This paper also contributes to the literature on asymptotic relative efficiency. Hodges and Lehmann (1956) was the first paper to propose a relative efficiency measure for testing, and they derived the efficiency of the t-test when the observations are normally distributed. Recently, He and Shao (1996) derived the Bahadur efficiency of closely-related normalized score tests. Their results show that t-tests are reasonably robust to heavy tails when considering the behavior of p-values. More recent work such as Canay and Otsu (2012) shows that many standard test statistics used in econometrics have these efficiency properties, when the moment generating function exists. Our results show that heavy tails lead to slow convergence of the Type-II error rate to zero, and therefore provide a new and different perspective on the efficiency properties of test-statistics. In particular, we show that the assumption in Canay and Otsu (2012) that the sample space is compact is an important assumption when considering the efficiency properties of the GEL test statistics.

For this paper we focus on the cases where we have an i.i.d. sample $X_1, \dots, X_n$. We are interested in inference on a parameter θ , and have a statistic nT_n which is asymptotically

chi-square locally to the null hypothesis that $\theta = \theta_0 + \delta/\sqrt{n}$:

$$nT_n \stackrel{\theta=\theta_0+\delta/\sqrt{n}}{=} \left\| \frac{1}{\sqrt{n}} \sum_{i=1}^n \psi_{\theta_0}(X_i) \right\|^2 + o_P(1) \Rightarrow \chi_r^2(\xi^2(\delta))$$

where $\xi^2(\delta)$ is typically a quadratic function in δ which depends on the derivatives of $\mathbb{E} \psi_{\theta_0}(X_i)$ in θ at θ_0 . Examples include standard Wald and Score tests derived from GEL procedures.

Under fixed alternatives, the behaviour of these test statistics can vary quite a bit, even among the class of tests which have the same limit distribution under local alternatives. The high-level assumption we will make here is that there exists a rate r_n and noncentrality parameter τ such that under a fixed alternative $\theta \neq \theta_0$,

$$r_n(T_n - \tau) \stackrel{\theta \neq \theta_0}{=} \frac{r_n}{n} \sum_{i=1}^n \tilde{\psi}(X_i) + o_P(1) \Rightarrow \mathcal{Z}$$

where in many cases $\mathcal{Z}$ is normal, but in some cases of practical interest $\mathcal{Z}$ can be alpha-stable.

This paper proceeds as follows. In Section 2, we present our main result on the converges rates when $\tilde{\psi}(X_i)$ exhibits heavy tails. In Section 3 we discuss how to construct robust statistics. In Section 4 we provide some simulation evidence. In Section 5 we demonstrate the utility of the new procedure in an application in asset pricing. In Section 6 we conclude. All proofs are available in the Appendix.

2 Large Deviation Results

In this section we present our main negative results on the large-deviations behaviour of asymptotically linear and self-normalised statistics when the data exhibit heavy-tails. We apply these results to the contraction rate of estimators and the asymptotic power of test-statistics.

The hypothesis we are interested in testing is

$$H_0 : a(\theta) = 0. \tag{2.1}$$

where $a : \mathbb{R}^p \rightarrow \mathbb{R}^r$ is a continuously differentiable function such that $A_0 := \frac{\partial}{\partial \theta} a(\theta_0)$ has rank $r \leq p$. We first introduce our general setting and the assumptions we will require for our results. We will consider i.i.d. observations $X_i \in \mathbb{R}^k$. We will assume that the parameter of

interest θ is identified as an M-estimator: there exists a known function $m(X_i, \theta)$ such that

$$\theta_0 = \arg \min_{\theta} \mathbb{E} m(X_i, \theta)$$

We can alternatively consider θ_0 as the unique zero of an L_2 derivative of m with respect to θ . Towards this direction, we will further assume that there exists a function g such that $\mathbb{E} g(X_i, \theta_0) = 0$ and

$$\mathbb{E}(m(X_i, \theta_0 + t) - m(X_i, \theta_0) - t'g(X_i, \theta_0))^2 = o(\|t\|^2) \quad (2.2)$$

for t in a neighbourhood around 0. This setup could be generalized, but as things stand it is in line with the literature on robustness to outliers going back to [Huber \(1964\)](#). We will make a basic assumption on g that the family of functions $\mathcal{G} := \{g(\cdot, \theta) : \theta \in \Theta\}$ satisfy uniform laws of large numbers and central limit theorems; such requirements are among the most common sufficient conditions provided for M-estimators to be consistent and asymptotically normal. It will also provide us with the means to verify our positive result on large-deviation properties of our new robust test later on. We briefly adopt the notation from [van der Vaart and Wellner \(2023\)](#) for uniform entropy $\sup_{\mathcal{Q}} N(\varepsilon, \mathcal{G}, L_2(\mathcal{Q}))$ and bracketing entropy $N_{[]}(\varepsilon, \mathcal{G}, L_2(P))$; for more discussion and definitions, see [Andrews \(1994\)](#) or [van der Vaart and Wellner \(2023\)](#), Section 2.

Assumption 2.1. *Let $\sup_{\mathcal{Q}} N(\varepsilon, \mathcal{G}, L_2(\mathcal{Q}))$ be the covering number for the class $\mathcal{G}$ and $N_{[]}(\varepsilon, \mathcal{G}, L_2(P))$ be the bracketing number for the class $\mathcal{G}$. Let (2.2) be satisfied for at $g(X_i, \theta_0) \in \mathcal{G}$. Then, there exist constants K and L such that either*

$$\begin{aligned} \sup_{\mathcal{Q}} N(\varepsilon, \mathcal{G}, L_2(\mathcal{Q})) &\leq \left(\frac{K}{\varepsilon}\right)^L, & \forall 0 < \varepsilon < K \\ N_{[]}(\varepsilon, \mathcal{G}, L_2(P)) &\leq \left(\frac{K}{\varepsilon}\right)^L, & \forall 0 < \varepsilon < K \end{aligned}$$

In particular, if g satisfies a stochastic differentiability property in θ , such that $\frac{\partial}{\partial \theta} \mathbb{E} g(X_i, \theta)$ exists and is bounded over Θ , such a class $\mathcal{G}$ will satisfy the above entropy bounds. Making these specific assumptions, rather than a higher-level Donsker-type assumption, is that these assumptions allow for use of exponential tail bounds on the empirical process in proving our positive large deviation result in Section 3. For our purposes, if our $m(X_i, \theta) = m(y_i - x_i' \theta)$, m convex, then this assumption will be satisfied. Another set of examples worth considering include when $g = \varphi \circ h$, where h satisfies the above entropy conditions and φ is monotone increasing with bounded derivative (in expectation) over the parameter space Θ .

To test (2.1) the most common test statistics are quadratic forms:

$$T_n := S_n' \hat{V}_n^{-1} S_n \quad (2.3)$$

where $\sqrt{n}S_n \Rightarrow \mathcal{N}(0, V_0)$ under the null hypothesis and $\hat{V}_n \xrightarrow{P} V_0$. We will also generally assume that the structure of V_0 and $\hat{V}_n$ is the general sandwich form, $\hat{V}_n = \hat{Q}' \hat{\Omega}^{-1} \hat{Q}$, and $V_0 = G_0' \Omega_0^{-1} G_0$. In our setup $G(\theta)' = \frac{\partial}{\partial \theta} \mathbb{E} g(X_i, \theta)$ and $\Omega(\theta) = \mathbb{E} g(X_i, \theta) g(X_i, \theta)'$, and we denote $G_0 = G(\theta_0)$, $\Omega_0 = \Omega(\theta_0)$. Our main high-level assumption on this statistic will be that it is asymptotically linear under fixed alternatives, and asymptotically chi-square under local alternatives. We can separate the quadratic form into a part that depends on the statistic S_n and a part that relies on the variance estimator. Under a fixed alternative with sufficient existence of moments, we expect that

$$\begin{aligned} S_n &\xrightarrow{P} \Delta \\ \hat{Q} &\xrightarrow{P} Q \\ \hat{\Omega} &\xrightarrow{P} \Omega \end{aligned}$$

where Δ is a function of θ_0 and θ_0 which depends on the structure of S_n , and in general $V_0 \neq V_1 := Q' \Omega^{-1} Q$. We can then consider a first-order expansion of the statistic around its probability limit: defining $\ell := \Omega^{-1} Q \Delta$, we have

$$S_n' \hat{V}_n^{-1} S_n - \Delta' V_1^{-1} \Delta = 2\ell' Q(S_n - \Delta) + 2\ell'(\hat{Q} - Q)\Delta - \ell'(\hat{\Omega} - \Omega)\ell + o_P(1) \quad (2.4)$$

Our main high-level assumption is that this expression, properly normalized, is asymptotically linear with a (nondegenerate) limit distribution.

Assumption 2.2. *For the class of test-statistics T_n defined in (2.3), there exists a noncentrality parameter $\tau := \Delta' Q' \Omega^{-1} Q \Delta$ and rate r_n such that $r_n/n \rightarrow 0$ and*

$$r_n(T_n - \tau) \stackrel{\theta \neq \theta_0}{\underset{n}{\rightrightarrows}} \frac{r_n}{n} \sum_{i=1}^n \tilde{\psi}(X_i) + o_P(1) \Rightarrow \mathcal{Z} \quad (2.5)$$

for some non-degenerate random variable $\mathcal{Z}$ such that there exists $\bar{\epsilon}$ such that $P(\mathcal{Z} < -\epsilon) > 0$ for all $0 < \epsilon < \bar{\epsilon}$. Furthermore, along the sequence $r(\theta_0) = \delta/\sqrt{n}$, we assume that

$$nT_n \stackrel{r(\theta_0) = \delta/\sqrt{n}}{\rightrightarrows} \chi_r^2(\xi^2(\delta)) \quad (2.6)$$

where $\xi^2(\delta) = \delta'(A_0(G'_0)^{-1}\Omega_0 G_0^{-1}A'_0)^{-1}\delta$.

This high-level assumption allows for applications to testing and estimation problems in most settings, as long as the ‘estimators’ S_n , $\hat{Q}$, and $\hat{\Omega}$ satisfy some regularity conditions. In particular, Wald test-statistics and Rao Score statistics satisfy these assumptions under standard regularity conditions. The first part of the assumption in (2.5) requires that the test-statistic, appropriately recentered and rescaled, is asymptotically linear under a fixed alternative. When $\mathbb{E}\|g(X_i, \theta^*)\|^4$, this assumption will generally be satisfied. It also can be satisfied when $\mathbb{E}\|g(X_i, \theta^*)\|^{2+\epsilon} = \infty$ for some $\epsilon \in (0, 2]$, in which case $\mathcal{Z}$ is α -stable, as long as $P(-\tilde{\psi}(X_i) > x)$ satisfies some regularity conditions; see e.g. Bentkus et al. (2007). Our result allows for only finitely many moments to exist, this will be relevant in our discussion of the t-test with infinite kurtosis. As discussed in Bentkus et al. (2007) the t-statistic is asymptotically linear under a fixed alternative, even when the variance is not asymptotically normal. This first assumption also covers the case where the data come from a thin-tailed distribution, which is ruled out by our second assumption below.

The assumption that $nT_n \Rightarrow \chi_r^2(\xi^2(\delta))$ in (2.6) is essentially an optimality condition on the sequence of tests using T_n under local alternatives; see Newey and McFadden (1994) for the construction of tests satisfying (2.6) and Romano et al. (2010) for a discussion of local asymptotically maxmin tests. The aim of this paper is to propose a new class of hypothesis tests that will improve the asymptotic behaviour of the hypothesis test under fixed alternatives while maintaining any optimality properties under a sequence of local alternatives. Thus, it is natural to assume that the tests we are comparing have desirable local power properties.

Our assumptions on heavy-tails are high-level assumptions on the tails of $-\tilde{\psi}(X_i)$. A function $F : \mathbb{R} \rightarrow \mathbb{R}$ is *regularly varying* if there exists $\gamma \in \mathbb{R}$ such that

$$\lim_{x \rightarrow \infty} \frac{F(\kappa x)}{F(x)} = \kappa^\gamma$$

for all $\kappa > 0$. In such a case, we say $F \in \mathcal{R}_\gamma$. Our second assumption characterises the class of distribution tails we consider in this paper.

Assumption 2.3. For $\gamma \in [0, 1)$, $\log P(-\tilde{\psi}(X_i) > y) \in \mathcal{R}_\gamma$.

This assumption implies that the tails of the distribution of $\tilde{\psi}(X_i)$ are heavier than those of an exponential random variable. In the latter case, we would use classic large-deviations theory (see e.g. Dembo and Zeitouni (2009)) using exponential-tilting and moment generating

functions. In fact, Assumption 2.3 precludes $\mathbb{E} e^{t\tilde{\psi}(X_i)}$ being finite for t in a neighborhood of 0. When we consider $\tilde{\psi}(X_i)$ in the context of the asymptotic expansion in (2.4), the (relative) convergence rates of S_n , $\hat{Q}$, and $\hat{\Omega}$ dictate whether all three terms or fewer are relevant for the asymptotic behaviour. Furthermore, the joint distribution of the three components (the two components of the sandwich variance estimator and the asymptotically linear ‘estimator’ S_n) are important for determining the tail heaviness.

This assumption relates to our results below in a direct sense: in our setting, the relevant rate-function for normalizing the log-Type-II error probabilities is

$$\Lambda(n) = -\log(nP(-\tilde{\psi}(X_i) > n))$$

This implies that $\Lambda(n)$ is concave in n for large n . This contrasts with the case for thin-tailed random variables where these log probabilities are asymptotically linear in n . Many kinds of tail behaviour seen in applications are included in our setting, such as Weibull, lognormal, Student-t, and Pareto tails. The assumption could be weakened to allow for oscillating tail behaviour for large y , and in this case could be replaced by a regularly-varying bound on the log-tail probabilities, however for ease of exposition we use Assumption 2.3.

2.1 Slow Convergence of Type-II error

With these assumptions in hand, we can state the main negative result on the asymptotic behaviour of $P(nT_n < C_\alpha)$ under large-deviations.

Theorem 1. *Suppose that Assumptions 2.2 and 2.3 are satisfied, with τ as in (2.5). Then, for the critical value C_α such that $P(nT_n > C_\alpha) \xrightarrow{\theta=\theta_0} \alpha$, we have that when $\theta = \theta^* \neq \theta_0$,*

$$\lim_{n \rightarrow \infty} \frac{-\log P(nT_n < C_\alpha)}{\Lambda(n)} \leq \tau^\gamma, \quad \gamma \in [0, 1).$$

Remark 1. We assume here that the observations are i.i.d. This is sufficient for such a bound to hold, but not necessary. The assumption of i.i.d. assumptions is used here in two places in the proof. In one place, we rewrite the probability than any two observations exceed a threshold as the product of the probabilities. In the other case we use that a given observation is independent of the other $n - 1$ observations to also write a joint probability as a product. Different assumptions on weak dependence would also allow us to rewrite such joint probabilities as products, which would provide a route to extending our results to the

case when the data are weakly-dependent.

The nature of the theorem is in terms of log-probabilities, which in the large deviations literature for heavy-tailed data are often referred to as “rough” large deviations. These results contrast with “precise” large deviations, which would directly approximate $P(nT_n < x)$ for large x on the probability scale. This literature is primarily focused on finding ranges of x in which the probabilities are best approximated by extreme value theory rather than a central limit theorem. For a discussion, see [Mikosch and Nagaev \(1998\)](#) and the references therein. While these results might be more useful from an approximation perspective, theoretically these approximations are much more sensitive to the behaviour of x and $\tilde{\psi}(X_i)$. They also require different tools depending on whether $\gamma > 0$ or $\gamma = 0$ in the result above. Here, we unify these cases under one result. Our results are also more useful from a statistical perspective, in that they relate to common measures of the relative efficiency of tests and estimators, which we now discuss.

From a statistical perspective, the log-probabilities are generally also the most relevant. Recall that for two probability measures on $\mathcal{X}$, the Kullback-Liebler divergence is defined as

$$K(Q, P) := \int_{\mathcal{X}} \log(dQ/dP) dQ$$

whenever Q is absolutely continuous with respect to P , and defined as taking on the value ∞ otherwise. Suppose we have a family of probability measures $\mathcal{P}$, and we would like to test whether the data-generating process $P \in \mathcal{P}_0 \subset \mathcal{P}$ against the alternative $H_1 : P \notin \mathcal{P}_0$. Then it is well-known that for a test W_n which is asymptotically level- α and rejects for $W_n > C_\alpha$, we have that under a fixed alternative $P_1 \in \mathcal{P} \setminus \mathcal{P}_0$,

$$\liminf_{n \rightarrow \infty} -\frac{1}{n} \log P(W_n < C_\alpha) \leq \inf_{P \in \mathcal{P}_0} K(P, P_1).$$

If a test achieves the righthand side of this inequality, the test is said to be Hodges-Lehmann efficient (see [Hodges and Lehmann \(1956\)](#)). In this sense, the Kullback-Leibler divergence characterises the fastest rate at which the probability of a type-II error converges to 0 under a fixed alternative. More recently, [Canay and Otsu \(2012\)](#) showed that for tests based on GEL have this efficiency property under the condition that the support of the observations is compact. Our theorem, applied to test statistics below, shows that in general efficiency of these procedures, in the Hodges-Lehmann sense, cannot be guaranteed when the support of the observations is not compact.

The original discussion in [Hodges and Lehmann \(1956\)](#) (see also [Serfling \(2009\)](#)) linked the asymptotic behaviour of log-probabilities to relative sample sizes. Consider two hypoth-

esis tests, which reject $H_0 : P \in \mathcal{P}_0$ for $T_{1n} > C_\alpha^1$ and $T_{2n} > C_\alpha^2$ respectively. Suppose each test is level- α . For a fixed alternative $P_1 \notin \mathcal{P}_0$, let $h(n)$ be the minimum sample size such that if $P(T_{1n} < C_\alpha^1) = \beta$, then $P(T_{2n} < C_\alpha^2) \leq \beta$. Then, we have that

$$\lim_{n \rightarrow \infty} \frac{h(n)}{n} = \lim_{n \rightarrow \infty} \frac{-\log P(W_{2n} < C_\alpha^2)}{-\log P(W_{1n} < C_\alpha^1)}$$

under P_1 if the limit exists. Thus, characterizing the asymptotic behaviour of log-probabilities is equivalent to characterizing the relative sample size needed when running a test using T_{1n} as a test statistic to have the same power against a fixed alternative as using T_{2n} . If $h(n) > n$ asymptotically, then eventually the marginal benefit from an additional observation when using T_{1n} is smaller than when using T_{2n} . Thus, for large sample sizes, this provides a guarantee that testing our hypothesis using T_{2n} is a more efficient use of the data relative to using T_{1n} .

2.2 Example: the t-test

Here we apply Theorem 1 to the 1-sample t-test for testing $\mathbb{E} X_i = 0$,

$$t_n^2 = \frac{n\bar{X}^2}{S^2}, \quad \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \quad S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Under the assumption that $\mathbb{E} X_i^2 < \infty$, $t_n^2 \Rightarrow \chi_1^2$ when $\mathbb{E} X_i = 0$. It is well-known (see for example [Newey and McFadden \(1994\)](#)) that the t-test has a minimax efficiency property under local-alternatives, i.e. when $\mathbb{E} X_i = \delta/\sqrt{n}$. In the setup of [Canay and Otsu \(2012\)](#), it is also clear that as a GEL test statistic, the t-test is optimal in a Hodges-Lehmann sense: if $\mathbb{E} X_i = \mu$ with $\mu^2 > 0$ for all n ,

$$\lim_{n \rightarrow \infty} -\frac{1}{n} \log P(t_n^2 < C_\alpha) = \inf_{P \in \mathcal{P}_\epsilon} K(P, P_1)$$

where $\mathcal{P}_\epsilon$ is the set of all distributions such that $\text{Var}(X_i) \geq \epsilon$ and C_α is the $1 - \alpha$ quantile of a χ_1^2 random variable. The setting of [Canay and Otsu \(2012\)](#) requires that the support of X_i is compact, which rules out any heavy-tailed random variables. If we apply our Theorem 1 to this setting, we need to first understand the weak convergence properties of t_n^2 under the fixed alternative. To that goal, expanding t_n^2 gives us

$$t_n^2 = \frac{n\bar{X}^2}{S^2} = \frac{n(\bar{X} - \mu)^2}{S^2} + \frac{2n(\bar{X} - \mu)\mu}{S^2} + \frac{n\mu^2}{S^2}.$$

Without loss of generality, we will assume that for these calculations $\text{Var}(X_i) = 1$. Then, the appropriate centring is $t_n^2 - n\mu^2$. [Bentkus et al. \(2007\)](#) showed that in this case, ruling out degenerate limits of S^2 , we have that

$$\frac{1}{\sqrt{n}}(t_n^2 - n\mu^2) = 2\mu \frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - \mu) - \mu^2 \frac{1}{\sqrt{n}} \sum_{i=1}^n ((X_i - \mu)^2 - 1) + o_P(1)$$

Theorem 1 depends then on

$$\tilde{\psi}(X_i) = 2\mu(X_i - \mu) - \mu^2((X_i - \mu)^2 - 1), \quad Q = 1, \quad \Omega = 1, \quad \Delta = \mu$$

Thus, to apply Theorem 1, the important contribution to the rate function $\Lambda(n)$ is the behaviour of

$$P(-2\mu(X_i - \mu) + \mu^2((X_i - \mu)^2 - 1) > x)$$

for large x . When X_i is lognormal, for example we have that

$$\frac{-\log P(-2\mu(X_i - \mu) + \mu^2((X_i - \mu)^2 - 1) > n)}{\log^2(n) - \log n} \rightarrow 1$$

When X_i follows a t-distribution with ν degrees of freedom, we then have

$$\frac{-\log P(-2\mu(X_i - \mu) + \mu^2((X_i - \mu)^2 - 1) > n)}{(\nu - 1) \log n} \rightarrow 1$$

In applying the results to the t-test, we show that the rate at which power converges to 1 (or Type-II error converges to 0) is typically not at an exponential rate when the data exhibit heavy tails. If the distribution only has finitely many moments but the mean is finite, power will still converge to 1, however it will converge at best at a polynomial rate.

Another way of connecting our Theorem 1 is that it complements work such as [Müller \(2023\)](#). In that paper, the worst-case behaviour of the t-statistic under the null is characterised. Here we characterise the worst-case behaviour of the t-test under fixed alternatives. In fact, if we consider an expanded class of distributions from [Müller \(2023\)](#), we can show that Theorem 1 is sharp in this class:

Proposition 1. *Let X_i^2 have a regularly-varying distribution function: $P(X_i^2 > x) = \ell(x)x^{-\nu/2}$ for $\nu > 1$, and let $\mathbb{E} X_i = \mu \neq 0$. Then,*

$$\lim_{n \rightarrow \infty} \frac{-\log P(t_n^2 < C_\alpha)}{(\nu - 1) \log n} = 1$$

The proof follows the proof of Theorem 3.3 in [Cline and Hsing \(2022\)](#). The $\nu - 1$ term in the denominator matches with intuition about consistency of the t-test: the t-test is a consistent test for the mean, as long as the mean exists, i.e. $\nu > 1$; for $\nu < 1$ generally the probability of a Type-II error, $P(t_n^2 < C_\alpha)$, will not converge to zero.

3 Robust GEL Statistics

We now turn towards our testing procedure that allows us to maintain the local efficiency properties of the class of tests T_n in (2.3) while incorporating a faster convergence rate of the Type-II error to zero than the rate specified in Theorem 1. Our proposed test statistic is constructed using GEL. In a recent paper, [Schennach and Wahlstrom \(2024\)](#) introduces the class of bounded-tilting estimators. These estimators form a subset of GEL estimators with good robustness properties. Recall that a GEL estimator for the parameter θ given moment function $g(X_i, \theta)$ is given by

$$\min_{\pi, \theta} \frac{1}{n} \sum_{i=1}^n h(n\pi_i) \quad (3.1)$$

$$s.t. \quad \sum_{i=1}^n \pi_i = 1 \quad (3.2)$$

$$\sum_{i=1}^n \pi_i g(X_i, \theta) = 0 \quad (3.3)$$

where h is a convex function. This $n + p$ dimensional problem is usually solved by finding the solution to the dual problem:

$$\min_{\theta} \max_{\eta, \lambda} \left\{ \eta - \frac{1}{n} \sum_{i=1}^n f(\eta + \lambda' g(X_i, \theta)) \right\} \quad (3.4)$$

where $f(y) = \sup_x \{xy - h(x)\}$ is the convex conjugate of h . Classic choices for h , and consequently f , are given in Table 1.

Procedure	$h(x)$	$f(y)$
Empirical Likelihood	$-\log x$	$-\log(1 - y)$
Exponential Tilting	$x \log x$	$\exp(y)$
Continuously Updating	x^2	$\frac{(1+y)^2}{2}$

Table 1: Common choices of h and f in GEL

In [Newey and Smith \(2004\)](#) it is suggested that Empirical Likelihood (EL) is a desirable

choice given that it has desirable higher-order efficiency properties, including smaller higher-order bias. Unfortunately, under misspecification the solution of (3.4) will only match with the solution to (3.1) asymptotically when the moment function g is bounded uniformly in θ ; see Schennach (2007). Other choices of h such as $x \log x$ for the Exponential Tilting (ET) estimator require the existence of higher moments for consistency and asymptotic normality. This motivated the introduction of bounded GEL estimators in Schennach and Wahlstrom (2024). These estimators use an f such that the support of f is $\mathbb{R}$ and the first derivative f' is uniformly bounded. This approach removed the necessity of requiring more than $\mathbb{E} \|g(X_i, \theta)\|^2 < \infty$ for asymptotic normality of the resulting estimator. Furthermore, the choice of f such that f is three-times continuously differentiable with $f'(0) = f''(0) = 1$ and $f'''(0) = 2$. These choices imply that the resulting estimator $\hat{\theta}_f$ will enjoy the same higher-order efficiency properties as the EL estimator, without requiring uniform boundedness of the moment function g .

We propose an alternative motivation for how to choose f based on our goal of improving the asymptotic relative efficiency of tests. Under the null hypothesis, we can estimate the Lagrange multipliers in (3.4) by solving a convex optimization problem:

$$(\hat{\eta}, \hat{\lambda}) = \arg \min_{\eta, \lambda} \left\{ \frac{1}{n} \sum_{i=1}^n f(\eta + \lambda' g(X_i, \bar{\theta})) - \eta \right\} \quad (3.5)$$

where $\bar{\theta}$ is any estimator of θ such that whenever $a(\theta_0) = \delta/\sqrt{n}$,

$$a(\bar{\theta}) = 0 \quad \forall n \quad (3.6)$$

$$\sqrt{n}(\bar{\theta} - \theta_0) = -B_0^{-1} A_0' (A_0 B_0^{-1} A_0')^{-1} \delta - B_0^{-1/2} M_0 B_0^{-1/2} G_0' \Omega^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n g(X_i, \theta_0) \quad (3.7)$$

$$B_0 = G_0' \Omega_0^{-1} G_0. \quad (3.8)$$

For construction of such an estimator, see Newey and McFadden (1994). In particular, any constrained minimizer or GEL estimator will satisfy these conditions.

It is well known (see e.g. Newey and Smith (2004)) that under the null hypothesis, $\sqrt{n}\hat{\lambda} = -\Omega_0^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n g(X_i, \theta_0) - \Omega^{-1} G_0 \sqrt{n}(\bar{\theta} - \theta_0) + o_P(1)$. In the literature it has been established that $\hat{\eta} = o_P(1/\sqrt{n})$; see e.g. Ragusa (2011) and Schennach and Wahlstrom (2024). In fact, with the choice of f such that $f'''(0) = 2$, it turns out that $\hat{\eta} = o_P(1/n)$. Constructing a test statistic in this case is therefore challenging, and in general the limit distribution of $\hat{\eta}$ will involve higher powers of the normal distribution and require the existence of higher-order moments.

As an alternative, we propose to use the bounded-tilting construction of [Schennach and Wahlstrom \(2024\)](#), with the following restrictions on f :

$$\begin{aligned} f_0 &:= f(0) = 0 \\ f_1 &:= f'(0) = 1 \\ f_2 &:= f''(0) = 1 \\ f_3 &:= f'''(0) = 0 \end{aligned}$$

An example of a function which satisfies these properties is the integrated Logistic CDF

$$f(y) = \log(1 + e^{2y}) - \log 2$$

where $f(y)$ has been scaled to satisfy $f_0 = f_3 = 0$ and $f_1 = f_2 = 1$. In our simulations and application below this is the choice of f we use, but our theoretical results do not rely on this particular choice. Our first positive result establishes the limit distribution of $\hat{\eta}$ for this class of functions f .

Theorem 2. *Let f be such that $f_0 = f_3 = 0$ and $f_1 = f_2 = 1$, and $\hat{\eta}$ as defined in (3.5), with $\bar{\theta}$ satisfying (3.6) and (3.7). Then, under $a(\theta_0) = \delta/\sqrt{n}$, we have that*

$$n\hat{\eta} \Rightarrow \chi_r^2(\xi^2(\delta))$$

The result is straightforward to prove using standard results from [Newey and McFadden \(1994\)](#), [van der Vaart and Wellner \(2023\)](#). The main innovation is using the implicit function theorem to obtain a useful expression for $\hat{\eta}$ in terms of $\hat{\lambda}$. The problem is convex in η , therefore for each fixed λ , there will be a unique solution $\hat{\eta}(\lambda)$ which satisfies the first order condition:

$$\frac{1}{n} \sum_{i=1}^n f'(\hat{\eta} + \lambda' g(X_i, \bar{\theta})) - 1 = 0 \tag{3.9}$$

In addition, the Lagrange multipliers $\hat{\lambda}$ satisfies

$$\frac{1}{n} \sum_{i=1}^n f'(\hat{\eta} + \hat{\lambda}' g(X_i, \bar{\theta})) g(X_i, \bar{\theta}) = 0$$

As has been established elsewhere (e.g. [Newey and Smith \(2004\)](#))

$$\sqrt{n}\hat{\lambda} = -\Omega_0^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n g(X_i, \theta_0) - \Omega^{-1} G_0 \sqrt{n}(\bar{\theta} - \theta_0) + o_P(1)$$

under the sequence $a(\theta_0) = \delta/\sqrt{n}$. Using the implicit function theorem with (3.9) allows us to derive a second-order expansion for $\hat{\eta}$ in $\hat{\lambda}$. The relevant calculations reveal that

$$\hat{\eta} = \hat{\lambda}' \Omega_0 \hat{\lambda} (1 - \frac{1}{2} f_3) + o_P(1)$$

Thus, if $f_3 = 2$, as in the case with EL, $\hat{\eta} = o_P(1/n)$. However, in the case we are interested in, $\hat{\eta} = O_P(1/n)$ and $n\hat{\eta} \Rightarrow \chi_r^2(\xi^2(\delta))$, as desired.

Remark 2. It is worth noting that the higher-order bias property of EL is exactly the property which leads to $\hat{\eta} = o_P(1/n)$ in our setting. In [Newey and Smith \(2004\)](#) it is shown that the property $f_3 = 2$ is sufficient for a higher-order bias term involving the third moments of $g(X_i, \theta_0)$ to drop out.

Remark 3. Choices other than $f_3 = 0$ can lead to equivalent inference (up to first-order). If we let $f(y) = e^y$, as in the ET procedure, we obtain $2n\hat{\eta} \Rightarrow \chi_r^2(\xi^2(\delta))$.

Remark 4. The statistic $\hat{\eta}$ is self-normalized, in the sense that no variance estimation is required. This is similar to criterion-based tests, however notice that tests based on $\hat{\eta}$ do not have the same challenge as tests based on the criterion function where correct specification can often be required to end up with an asymptotically pivotal statistic.

Remark 5. We can extend this result to the time-series case, with the case of martingale difference sequences being straightforward. For the case of general mixing sequences, it is likely that the limit distribution of $\hat{\eta}$ will be non-pivotal. Thus, in this case a more promising direction would be to either modify the empirical likelihood objective function (see e.g. [Kitamura \(2007\)](#)) or employ simulation methods, such as the bootstrap or subsampling, to conduct inference.

We now present our result that $\hat{\eta}$ has desirable properties from a large deviations perspective, irrespective of how heavy the tails of $g(X_i, \theta)$ are. We assume that there exists $\theta^* \in \{\theta : a(\theta) = 0\}$ such that $\bar{\theta} \xrightarrow{P} \theta^*$.

Theorem 3. *We assume that 0 is in the convex hull of the support of $g(X_i, \theta^*)$ along a sequence such that there exists some $d > 0$ such that $\|a(\theta_0)\| \geq d$ for all n but $a(\theta^*) = 0$.*

Define

$$\Gamma(t) := -\inf_{\rho > 0} \{\log \mathbb{E} \exp\{\rho(f'(t + \lambda_0' g(X_i, \theta^*)) - 1)\}\}$$

Then, we have that

$$\lim_{n \rightarrow \infty} -\frac{1}{n} \log P(n\hat{\eta} < C_\alpha) \geq \Gamma(0) > 0.$$

Remark 6. The assumption on the convex hull of the support of $g(X_i, \theta^*)$ is necessary for the existence of the Lagrange multipliers η_0 and λ_0 . A classic reference on these existence results is [Borwein and Lewis \(1993\)](#).

Remark 7. The proof involves some empirical process methods, namely the fact that the function class $\eta + \lambda'y$ has finite VC dimension, and thus so does $f'(\eta + \lambda'y)$ since f' is monotone increasing. Boundedness of f' allows for approximating the large deviation behaviour of $\hat{\eta}$ with the idealized estimator

$$\tilde{\eta} = \arg \min_{\eta} \left\{ \frac{1}{n} \sum_{i=1}^n f(\eta + \lambda'_0 g(X_i, \theta^*)) - \eta \right\} \quad (3.10)$$

using an exponential bound on the probability that $|\hat{\eta} - \tilde{\eta}| > \epsilon$. Boundedness is sufficient for such a relationship to hold, but is not necessary. See [van der Vaart and Wellner \(2023\)](#), Chapter 2.14, for more details.

Remark 8. The function Γ is the well-known rate function for the scores $f'(\lambda'g(X_i, \theta^*)) - 1$. The expression can be understood as a measure of the distance between the null and alternative: when $\theta = \theta_0$, $\eta_0 = 0$ and $\lambda_0 = 0$, however under the alternative θ^* , $\eta_0 > 0$ and $\|\lambda_0\| > 0$. The score for $\tilde{\eta}$ is $f'(\eta + \lambda'_0 g(X_i, \theta^*)) - 1$. This function is mean-zero if and only if $\eta = \eta_0$, and thus the mean of the difference $f'(\lambda'_0 g(X_i, \theta^*)) - 1 = f'(\lambda'_0 g(X_i, \theta^*)) - f'(\eta_0 + \lambda'_0 g(X_i, \theta^*))$ is strictly negative; the larger this difference, on average, the higher power we should expect against the alternative in question.

Remark 9. For dependent data, the exponential tail bound that allows us to compare $\hat{\eta}$ to the infeasible $\tilde{\eta}$ defined in (3.10) could be replaced using a generalisation of Bernstein's inequality for mixing processes. Then, large deviation results for dependent data could be used to derive the rate function for $\tilde{\eta}$.

4 Simulations

We provide simulation results calibrated to arithmetic returns from the SPDR S&P 500 ETF Trust (SPY), as in [Shephard \(2020\)](#). We simulate data from the following DGP:

$$\begin{aligned} y_i &= (z_i - \psi)\beta_1 + \varepsilon_i \\ z_i &= \psi + V\sigma_z \\ \varepsilon_i &\sim \mathcal{N}(0, (1 + |z_i - \psi|^\zeta)C^2) \end{aligned} \tag{4.1}$$

We choose three different distributions for V :

$$\begin{aligned} \sqrt{\frac{\nu}{\nu-2}}V_\nu &\sim t_\nu, \quad \nu \in \{\nu_l, \nu_h\} \\ \exp\{-v/4\}V &\sim \mathcal{N}(0, \eta_i^{-1}), \quad \log \eta_i \sim \mathcal{N}(0, v), \quad v \in \{v_l, v_h\} \end{aligned}$$

where t_ν is a student-t distribution with ν degrees of freedom. Here, z_i are calibrated to match the weekly returns. We use data from 1st August 2018, through 18th November 2022. We chose this range for the simulation calibration because the first two years of data exhibit more volatility due to inclusion of the start of the global pandemic in spring of 2020. We use the first two years to estimate ν_h and v_h for our “higher volatility” setup. We used the full two years to calibrate ν_l and v_l . We set $\nu_h = 2.219$, $\nu_l = 3.352$, $v_l = 0.933$, and $v_h = 1.663$. We then set σ_Z so that the standard deviation matches the sample standard deviation of the returns during the relevant time period, and ψ is likewise the sample mean for each time period. We assume ψ is known, so that we can perform a heteroskedasticity-robust test of the null $H_0 : \beta_1 = 1$ using the t-statistic formed from the observations $X_i = (z_i - \psi)(y_i - (z_i - \psi)\beta_0)$. Unlike [Shephard \(2020\)](#), we include conditional heteroskedasticity. We do this so that the t-test is clearly a reasonable thing to do here, rather than a more efficient procedure that leverages homoskedasticity. The form of (4.1) is motivated so that for our choice of ζ , T_n provides asymptotically valid inference under the null, and we choose ζ to be 0.12 in the high-volatility case when z_i are student-t, and set $\zeta = 0.5$ in the lower-volatility setting. For the lognormal mixtures, we set $\zeta = 1.0$. C is chosen so that $\mathbb{E}\varepsilon_i^2 = 4$, as in [Shephard \(2020\)](#). We set the null value of β as $\beta_{1,0} = 1$. Our simulations will display the power of the tests as n increases, for a fixed alternative.

In Figure 1, we plot power curves as a function of sample size. All power calculations are performed after correcting the critical values so that the size of all tests is 0.05. Notice that the sample size required by the t-test to achieve a particular power level is uniformly

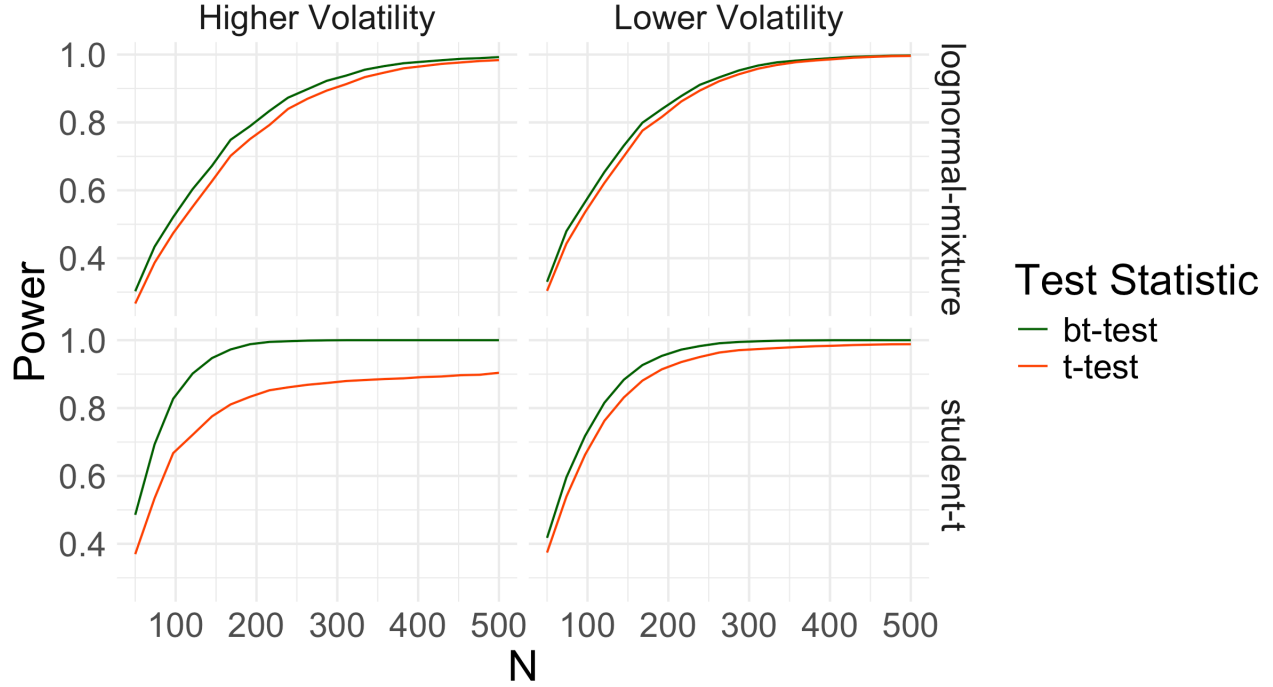


Figure 1: Comparison of New Test with t-test

This figure plots the power when testing $\beta_1 = 1$ against a two-sided alternative $\beta_1 \neq 1$. The power curves have been (point-wise in sample size) size-corrected. The orange line corresponds to the classic t-test, while the green line corresponds to the new GEL test.

larger than the sample size required by the new robust procedure. The new robust procedure is implemented using the moment condition $\mathbb{E} X_i = 0$, and $f(y) = \log(1 + e^{2y})$. The new test gets arbitrarily close to 1 for all data generating processes considered, as sample size increases, whereas the power curve for the traditional t-test flattens out considerably as the sample size gets large, especially when the data are student-t. This supports the claim in Theorem 3 that the new testing procedure leads to better Type-II error properties.

In Figure 2 we plot the achieved significance level as a function of sample size. Note that the power calculations in 1 are done after size-correcting, so the results are not driven by the slight over-rejection of the null for small sample sizes.

5 Application

As an application we use the SPDR S&P 500 ETF Trust (SPY) data from 1st August 2022 to 1st January 2025. We consider the regressions of the individual weekly arithmetic returns

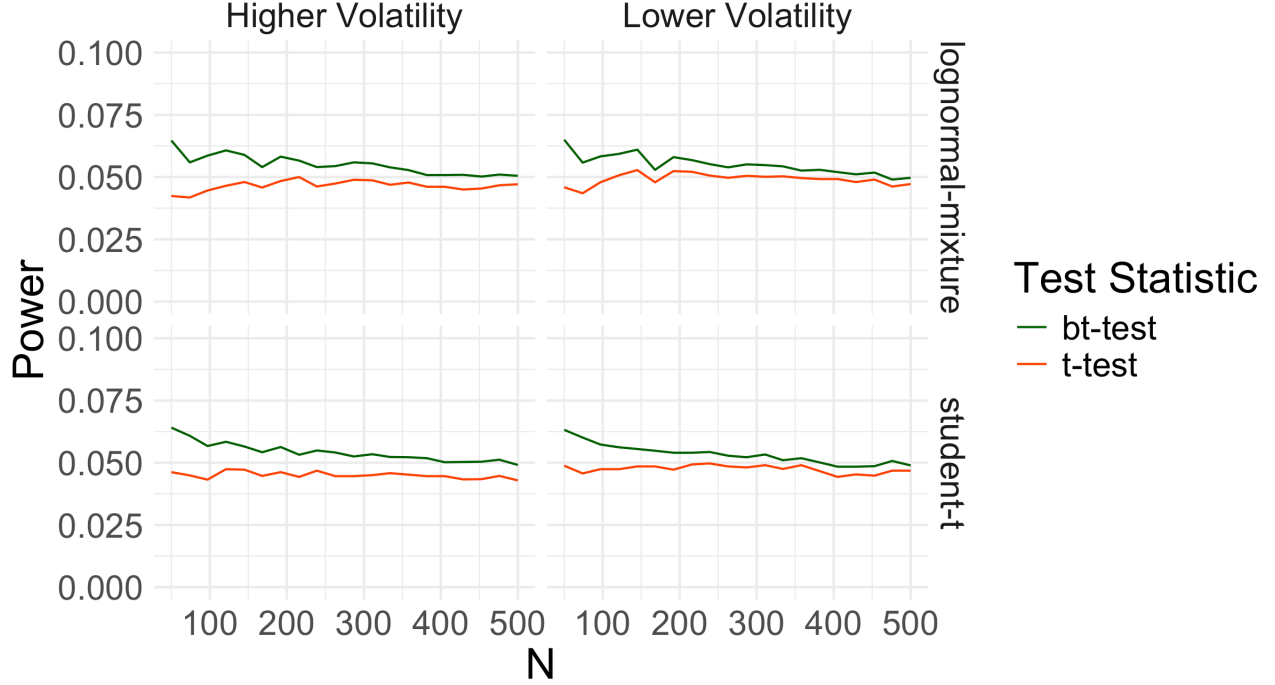


Figure 2: Comparison of New Test with t-test

This figure plots the size when testing $\beta_1 = 1$ against a two-sided alternative $\beta_1 \neq 1$. The orange line corresponds to the classic t-test, while the green line corresponds to the new GEL test.

y_i on the market index x_i . In particular, in the model

$$y_{i,t} = \alpha_i + (x_t - \mathbb{E} x_t)\beta_i + \varepsilon_{i,t}$$

we are interested in testing $H_0 : \beta_i = 1$ for all i . This test can be interpreted as testing whether particular betas carry more or less (nondiversifiable) risk than the market index. Given the multiple-testing problem, power becomes a concern for correctly grouping assets based on beta. The low-beta anomaly has long been of interest, going back to [Jensen et al. \(1972\)](#). Explaining the anomaly remains an area of active research, with papers such as [Andrei et al. \(2023\)](#) proposing alternative explanations of how investors interact with information beta. In other studies, such as [DeMiguel et al. \(2024\)](#), preliminary estimates of beta and volatility are used as inputs into a richer multifactor structure.

For our application here, we compare two testing procedures: our new testing procedure using the OLS first-order conditions as the moment conditions, under the assumption that $\beta_i = 1$, and the standard t-ratio test using the OLS estimate $\hat{\beta}$. We use robust standard errors in constructing the t-ratios. The regressions are performed as time-series regressions for each stock i .

The heavy-tailed environment is relevant here, based on estimates for the tail index of the scores, $(x_i - \mathbb{E} x_i)\varepsilon_i$. Using the residuals from least-squares estimation, the tail-index for 97% of stocks is below 3, implying that fewer than 3 moments for these scores exist.

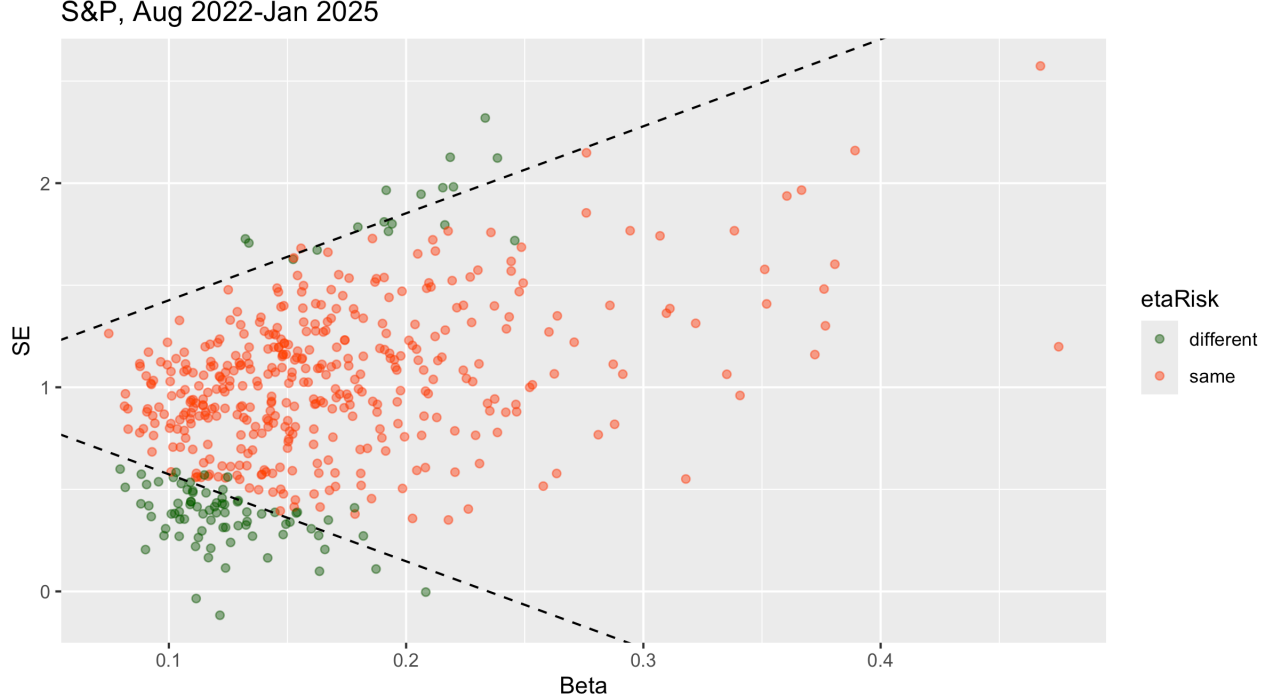


Figure 3: Classifying β

This figure plots the results from testing $\beta_1 = 1$ against a two-sided alternative $\beta_1 \neq 1$. The dashed lines show the rejection region of the standard t-test. The green points are those that classified as having a different (nondiversifiable) risk-level relative to the market using our new test, and the orange points represent those points which are classified as having the same risk level.

We implement our GEL-based test using $f(y) = \log(1 + e^{2y})$. In Figure 3 we see the results of our analysis. The t-test classifies 11 stocks as being ‘high-beta’ while the new test classifies 17. Similarly, t-test classifies 68 stocks as being ‘low-beta’ while the new test classifies 78. Size is corrected for multiple comparisons on 495 assets at the 0.05 level using a Bonferroni correction. The increased detection at such a low significance level demonstrates the desirable power properties of the new procedure.

6 Conclusion

This paper provides new results on the efficiency of tests when the data have heavy tails. These results are the first to demonstrate the theoretical inefficiency of the t-test with heavy-tailed data in an asymptotic setting. Our results complement recent work studying the

performance of test statistics under the null hypothesis when the data exhibit heavy tails. Our simulations demonstrate that the theoretical predictions match well with finite-sample behaviour. In the application we find that using a robust testing procedure accounts for higher detection rates of lower-risk and higher-risk assets.

It would be desirable to extend the main results to cases in which the dimension of the moment conditions are increasing with n . In such a case, a two-step procedure moments are selected in the first step seems necessary to prevent too much power loss. Given the extensive applications to finance, the potential for incorporating high dimensionality (e.g. conditional moment condition models) and serial dependence seems like it would be valuable for practitioners.

The new procedure is compatible with the bootstrap. With any $\sqrt{n}$ consistent estimate of $\hat{\theta}$, the bootstrap test statistics can be calculated as

$$(\hat{\eta}^*, \hat{\lambda}) = \arg \min_{\eta, \lambda} \frac{1}{n} \sum_{i=1}^n f(\eta + \lambda' g(X_i^*, \hat{\theta}))$$

and we can reject whenever $n\hat{\eta} > \bar{C}$ where $\bar{C}$ is chosen to be the $1 - \alpha$ quantile of the bootstrap distribution of $n\hat{\eta}^*$. It would be of interest to study whether the bootstrap offers asymptotic refinements.

References

- ANDREI, D., J. CUJEAN, AND M. WILSON (2023): “The lost capital asset pricing model,” *Review of Economic Studies*, 90, 2703–2762.
- ANDREWS, D. W. (1994): “Empirical process methods in econometrics,” *Handbook of econometrics*, 4, 2247–2294.
- BENTKUS, V., B. Y. JING, Q. M. SHAO, AND W. ZHOU (2007): “Limiting Distributions of the Non-Central t-Statistic and Their Applications to the Power of t-Tests under Non-Normality,” *Bernoulli*, 13, 346–364.
- BORWEIN, J. M. AND A. S. LEWIS (1993): “Partially-finite programming in L_1 and the existence of maximum entropy estimates,” *SIAM Journal on Optimization*, 3, 248–267.
- CANAY, I. A. AND T. OTSU (2012): “Hodges–Lehmann Optimality for Testing Moment Conditions,” *Journal of Econometrics*, 171, 45–53.
- CLINE, D. B. AND T. HSING (2022): “Large deviation probabilities for sums of random variables with heavy or subexponential tails,” *arXiv preprint arXiv:2211.16340*.
- DEMBO, A. AND O. ZEITOUNI (2009): *Large Deviations Techniques and Applications*, Stochastic Modelling and Applied Probability, Springer Berlin Heidelberg.
- DEMIGUEL, V., A. MARTÍN-UTRERA, AND R. UPPAL (2024): “A Multifactor Perspective on Volatility-Managed Portfolios,” *The Journal of Finance*, 79, 3859–3891.
- EFRON, B. (1969): “Student’s t-Test under Symmetry Conditions,” *Journal of the American Statistical Association*, 64, 1278–1302.
- GINÉ, E., F. GÖTZE, AND D. M. MASON (1997): “When Is the Student T-Statistic Asymptotically Standard Normal?” *The Annals of Probability*, 25, 1514–1531.
- HE, X. AND Q.-M. SHAO (1996): “Bahadur Efficiency and Robustness of Studentized Score Tests,” *Annals of the Institute of Statistical Mathematics*, 48, 295–314.
- HJORT, N. L. AND D. POLLARD (2011): “Asymptotics for Minimisers of Convex Processes,” *arXiv preprint arXiv:1107.3806*.
- HODGES, J. AND E. LEHMANN (1956): “The Efficiency of Some Nonparametric Competitors of the T-Test,” *Annals of Mathematical Statistics*, 27, 324–335.

- HUBER, P. J. (1964): “Robust Estimation of a Location Parameter,” *The Annals of Mathematical Statistics*, 35, 73 – 101.
- JENSEN, M. C., F. BLACK, AND M. S. SCHOLES (1972): “The capital asset pricing model: Some empirical tests,” .
- KITAMURA, Y. (2007): “Empirical likelihood methods in econometrics: theory and practice,” in *Advances in Economics and Econometrics: Theory and Applications, Ninth World Congress, 2007*, Cambridge University Press.
- LE CAM, L. (2012): *Asymptotic methods in statistical decision theory*, Springer Science & Business Media.
- LEHMANN, E. L. AND J. P. ROMANO (2005): *Testing Statistical Hypotheses*, vol. 3, Springer.
- MIKOSCH, T. AND A. V. NAGAEV (1998): “Large Deviations of Heavy-Tailed Sums with Applications in Insurance,” *Extremes. Statistical Theory and Applications in Science, Engineering and Economics*, 1, 81–110.
- MÜLLER, U. K. (2019): “Refining the Central Limit Theorem Approximation via Extreme Value Theory,” *Statistics & Probability Letters*, 155, 108564.
- (2023): “A more robust t-test,” *Review of Economics and Statistics*, 1–46.
- NEWKEY, W. K. AND D. MCFADDEN (1994): “Large Sample Estimation and Hypothesis Testing,” *Handbook of Econometrics*, 4, 2111–2245.
- NEWKEY, W. K. AND R. J. SMITH (2004): “Higher order properties of GMM and generalized empirical likelihood estimators,” *Econometrica*, 72, 219–255.
- PITMAN, E. J. (1949): “Notes on Non-Parametric Statistical Inference,” Tech. rep., North Carolina State University. Dept. of Statistics.
- RAGUSA, G. (2011): “Minimum divergence, generalized empirical likelihoods, and higher order expansions,” *Econometric Reviews*, 30, 406–456.
- ROMANO, J. P., A. M. SHAIKH, AND M. WOLF (2010): “Hypothesis testing in econometrics,” *Annu. Rev. Econ.*, 2, 75–104.
- SCHENNACH, S. M. (2007): “Point estimation with exponentially tilted empirical likelihood,” .

- SCHENNACH, S. M. AND O. WAHLSTROM (2024): “Bounded tilting estimation,” *Econometric Reviews*, 1–14.
- SERFLING, R. J. (2009): *Approximation Theorems of Mathematical Statistics*, vol. 162, John Wiley & Sons.
- SHAO, Q.-M. (1999): “A Cramér Type Large Deviation Result for Student’s t-Statistic,” *Journal of Theoretical Probability*, 12, 385–398.
- SHEPHARD, N. (2020): “An Estimator for Predictive Regression : Reliable Inference for Financial Economics,” Tech. rep., Harvard University.
- VAN DER VAART, A. AND J. A. WELLNER (2023): *Weak Convergence and Empirical Processes: With Applications to Statistics*, Springer.
- VAN DER VAART, A. W. (2000): *Asymptotic statistics*, vol. 3, Cambridge university press.
- YOUNG, A. (2022): “Consistency without inference: Instrumental variables in practical application,” *European Economic Review*, 147, 104112.

A Proofs of Main Results

In the appendix we will use the notation $\mathbb{P}_n f := \frac{1}{n} \sum_{i=1}^n f(X_i)$ and $\mathbb{P}_{n,-i} f := \frac{1}{n} \sum_{j \neq i}^n f(X_j)$, and $\mathbb{G}_n f = \frac{1}{\sqrt{n}} \sum_{i=1}^n f(X_i) - \mathbb{E} f(X_i)$.

A.1 Proof of Theorem 1

Proof. The proof of Theorem 1 is done by deriving a lower bound for the log tail probabilities.

First, note that there is some $R_n = o_P(1)$ such that $r_n(T_n - \tau) = r_n \mathbb{P}_n \tilde{\psi} + R_n$. This implies that

$$\begin{aligned} P(nT_n < C_\alpha) &= P(r_n(T_n - \tau) < -r_n\tau + C_\alpha r_n/n) \\ &= P\left(r_n \mathbb{P}_n \tilde{\psi} + R_n < -r_n\tau + C_\alpha r_n/n\right) \end{aligned}$$

By Lemma 4, we can conclude that along some sub-subsequence, for N large enough, $|R_{n_{k(l)}}| < \epsilon$ for all $n_{k(l)} \geq N$. The lemma allows us to replace R_n with ϵ , derive a lower bound for $P\left(r_n \mathbb{P}_n \tilde{\psi} < -r_n\tau + C_\alpha r_n/n\right)$, and if the limit exists (after normalised by $\Lambda(n)$), then we will have an upper bound for the normalised limit of $-\log P(nT_n < C_\alpha)$. Let $d_n := r_n\tau - C_\alpha/n$. Proceeding, we have that

$$\begin{aligned} &P\left(r_n \mathbb{P}_n \tilde{\psi} + \epsilon < -d_n\right) \\ &\geq P\left(-r_n \mathbb{P}_n \tilde{\psi} - \epsilon > d_n, \max_i \{-\tilde{\psi}(X_i)\} > n\tau\right) \\ &= P\left(-r_n \mathbb{P}_n \tilde{\psi} - \epsilon > d_n, \bigcup_i \left[-\tilde{\psi}(X_i) > n\tau\right]\right) \\ &\geq \sum_{i=1}^n \left(-r_n \mathbb{P}_n \tilde{\psi} - \epsilon > d_n, -\tilde{\psi}(X_i) > n\tau\right) \\ &\quad - \sum_{j < i} P\left(-r_n \mathbb{P}_n \tilde{\psi} + \epsilon > d_n, -\tilde{\psi}(X_i) > n\tau, -\tilde{\psi}(X_j) > n\tau\right) \\ &\geq \sum_{i=1}^n P\left(-r_n \mathbb{P}_{n,-i} \tilde{\psi} + r_n\tau - \epsilon > d_n, -\tilde{\psi}(X_i) > n\tau\right) \\ &\quad - n \left(\frac{n-1}{2}\right) P\left(-\tilde{\psi}(X_i) > d_n\right)^2 \\ &= nP\left(-\tilde{\psi}(X_i) > n\tau\right) \\ &\quad \cdot \left(P_{i,n}(\epsilon) - \left(\frac{n-1}{2}\right) P\left(-\tilde{\psi}(X_i) > n\tau\right)\right) \end{aligned}$$

where $P_{i,n}(\epsilon) := P\left(-r_n \mathbb{P}_{n,-i} \tilde{\psi} > \epsilon - C_\alpha/n\right)$. We have that

$$P_{i,n}(\epsilon) \rightarrow P(\mathcal{Z} < -\epsilon)$$

Since $nP(-\tilde{\psi}(X_i) > n\tau) \rightarrow 0$, we then have that

$$\begin{aligned} \frac{-\log\left(P_{i,n}(\epsilon) - \left(\frac{n-1}{2}\right) P(-\tilde{\psi}(X_i) > n\tau)\right)}{\Lambda(n)} &= \frac{-\log(P(\mathcal{Z} < -\epsilon) + o(1))}{\Lambda(n)} \\ &\rightarrow 0 \end{aligned}$$

for all $\epsilon > 0$, as $P(\mathcal{Z} < 0) > 0$. Therefore, we have that

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{-\log P(nT_n < C_\alpha)}{\Lambda_n(n)} &\leq \lim_{n \rightarrow \infty} \frac{\Lambda(n\tau) - \log\left(P_{i,n}(\epsilon) - \left(\frac{n-1}{2}\right) P(-\tilde{\psi}(X_i) > n\tau)\right)}{\Lambda(n)} \\ &= \tau^\gamma \end{aligned}$$

as desired. □

A.2 Proof of Theorem 2

Theorem 2 is proved by proving three lemmas: Lemma 1, 2, and 3. Following Newey and McFadden (1994), we define

$$\begin{aligned}
\Omega_0 &= \mathbb{E} g(X_i, \theta_0) g(X_i, \theta_0)' \\
G_0 &= \frac{\partial}{\partial \theta} \mathbb{E} g(X_i, \theta_0) \\
A_0 &= \frac{\partial}{\partial \theta} a(\theta_0) \\
B_0 &= G_0' \Omega_0^{-1} G_0 \\
M_0 &= I - B_0^{-1/2} A_0' (A_0 B_0^{-1} A_0)^{-1} A_0 B_0^{-1/2} \\
\mathbb{P}_n g_0 &= \frac{1}{n} \sum_{i=1}^n g(X_i, \theta_0) \\
\mathbb{P}_n \bar{g} &= \frac{1}{n} \sum_{i=1}^n g(X_i, \bar{\theta}) \\
\mathbb{G}_n g_0 &= \frac{1}{\sqrt{n}} \sum_{i=1}^n g(X_i, \theta_0) - \mathbb{E} g(X_i, \theta_0) \\
&= \frac{1}{\sqrt{n}} \sum_{i=1}^n g(X_i, \theta_0) \\
\mathbb{G}_n \bar{g} &= \frac{1}{\sqrt{n}} \sum_{i=1}^n g(X_i, \bar{\theta}) - \mathbb{E} g(X_i, \bar{\theta})
\end{aligned}$$

Lemma 1. *Let $\bar{\theta}$ be an estimator of θ such that $a(\bar{\theta}) = 0$ for all n , and for $a(\theta_0) = \delta/\sqrt{n}$,*

$$\sqrt{n}(\bar{\theta} - \theta_0) = -B_0^{-1} A_0' (A_0 B_0^{-1} A_0')^{-1} \delta - B_0^{-1/2} M_0 B_0^{-1/2} G_0' \Omega^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n g(X_i, \theta_0).$$

Then, we have that

$$\begin{aligned}
\sqrt{n} \hat{\lambda} &= -\Omega_0^{-1} \mathbb{G}_n \bar{g} + o_P(1) \\
&= -\Omega_0^{-1} (\mathbb{G}_n g_0 + G_0 \sqrt{n}(\bar{\theta} - \theta_0)) + o_P(1)
\end{aligned}$$

Proof. Consider the objective function

$$\frac{s}{\sqrt{n}} - \frac{1}{n} \sum_{i=1}^n f \left(\frac{s}{\sqrt{n}} + \frac{t'}{\sqrt{n}} g(X_i, \bar{\theta}) \right)$$

which has minimizers $(\sqrt{n}\hat{\eta}, \sqrt{n}\hat{\lambda})$. By the mean-value theorem we have that, recalling that $f(0) = 0$ and $f'(0) = 1$,

$$\frac{1}{n} \sum_{i=1}^n f \left(\frac{s}{\sqrt{n}} + \frac{t'}{\sqrt{n}} g(X_i, \bar{\theta}) \right) - \frac{s}{\sqrt{n}} \quad (\text{A.1})$$

$$= \frac{1}{\sqrt{n}} \mathbb{P}_n \bar{g}' t + \frac{1}{2} \frac{t'}{\sqrt{n}} \frac{1}{n} \sum_{i=1}^n f'' \left(\frac{\tilde{s}}{\sqrt{n}} + \frac{\tilde{t}'}{\sqrt{n}} g(X_i, \bar{\theta}) \right) g(X_i, \bar{\theta}) g(X_i, \bar{\theta})' \frac{t}{\sqrt{n}} \quad (\text{A.2})$$

$$+ \frac{s}{\sqrt{n}} \frac{1}{n} \sum_{i=1}^n f'' \left(\frac{\tilde{s}}{\sqrt{n}} + \frac{\tilde{t}'}{\sqrt{n}} g(X_i, \bar{\theta}) \right) g(X_i, \bar{\theta})' \frac{t}{\sqrt{n}} \quad (\text{A.3})$$

$$+ \frac{1}{2} \frac{1}{n} \sum_{i=1}^n f'' \left(\frac{\tilde{s}}{\sqrt{n}} + \frac{\tilde{t}'}{\sqrt{n}} g(X_i, \bar{\theta}) \right) \frac{s^2}{n} \quad (\text{A.4})$$

$$(\text{A.5})$$

Define

$$\begin{aligned} Q_n(s, t) &:= \sum_{i=1}^n f \left(\frac{s}{\sqrt{n}} + \frac{t'}{\sqrt{n}} g(X_i, \bar{\theta}) \right) - \sqrt{n}s \\ \tilde{Q}_n(s, t) &:= \mathbb{G}_n g'_0 t + \sqrt{n}(\bar{\theta} - \theta_0)' G'_0 t + \frac{1}{2} t' \Omega_0 t \\ &\quad + \frac{1}{\sqrt{n}} \mathbb{G}_n g'_0 t s + (\bar{\theta} - \theta_0)' G'_0 t s + \frac{1}{2} s^2 \end{aligned}$$

noting that the minimizers are

$$\arg \min_{s,t} Q_n(s, t) = (\sqrt{n}\hat{\eta}, \sqrt{n}\hat{\lambda})$$

$$\arg \min_t \tilde{Q}_n(s, t) = (\hat{s}, \hat{t})$$

$$\begin{aligned} \hat{t} &= -(\Omega_0 - \frac{1}{\sqrt{n}}(\mathbb{G}_n g_0 + G_0 \sqrt{n}(\bar{\theta} - \theta_0))(\mathbb{G}_n g_0 + G_0 \sqrt{n}(\bar{\theta} - \theta_0))')^{-1} \\ &\quad \cdot (\mathbb{G}_n g_0 + G_0 \sqrt{n}(\bar{\theta} - \theta_0)) \\ &= -\Omega_0^{-1}(\mathbb{G}_n g_0 + G_0 \sqrt{n}(\bar{\theta} - \theta_0)) + o_P(1) \\ \hat{s} &= \frac{1}{\sqrt{n}}(\mathbb{G}_n g_0 + G_0 \sqrt{n}(\bar{\theta} - \theta_0))' \Omega_0^{-1}(\mathbb{G}_n g_0 + G_0 \sqrt{n}(\bar{\theta} - \theta_0)) + o_P(1) \\ &= o_P(1) \end{aligned}$$

For each s, t , $|Q_n(s, t) - \tilde{Q}_n(s, t)| \xrightarrow{P} 0$. To see this, use the mean value expansion in (A.2)-(A.4), and consider the differences

$$\sqrt{n} \mathbb{P}_n \bar{g} - \mathbb{G}_n g_0 - G_0 \sqrt{n}(\bar{\theta} - \theta_0) \tag{A.6}$$

$$\frac{1}{n} \sum_{i=1}^n f'' \left(\frac{\tilde{s}}{\sqrt{n}} + \frac{\tilde{t}'}{\sqrt{n}} g(X_i, \bar{\theta}) \right) g(X_i, \bar{\theta}) g(X_i, \bar{\theta})' - \Omega_0 \tag{A.7}$$

since clearly $\frac{1}{n} \sum_{i=1}^n f'' \left(\frac{\tilde{s}}{\sqrt{n}} + \frac{\tilde{t}'}{\sqrt{n}} g(X_i, \bar{\theta}) \right) \xrightarrow{P} 1$. For (A.6), note that by recentering $\sqrt{n} \mathbb{P}_n \bar{g}$, we have

$$\begin{aligned} \sqrt{n} \mathbb{P}_n \bar{g} - \mathbb{G}_n g_0 - G_0 \sqrt{n}(\bar{\theta} - \theta_0) &= \mathbb{G}_n(\bar{g} - g_0) + \sqrt{n} \mathbb{E} g(X_i, \bar{\theta}) - G_0 \sqrt{n}(\bar{\theta} - \theta_0) \\ &= \mathbb{G}_n(\bar{g} - g_0) + o_P(1) \\ &= o_P(1) \end{aligned}$$

using the delta-method and the differentiability of $\mathbb{E} g(X_i, \theta)$ at θ_0 and the stochastic equicontinuity condition on the family of g indexed by θ . Thus, the minimizer of $Q_n(s, t)$ is close the the minimizer of $\tilde{Q}_n(s, t)$, up to possibly $o_P(1)$ terms; see Hjort and Pollard (2011), Lemma 1 and Lemma 2. This implies that

$$\begin{aligned} \sqrt{n}\hat{\eta} &= o_P(1) \\ \sqrt{n}\hat{\lambda} &= -\Omega_0^{-1}(\mathbb{G}_n g_0 + G_0 \sqrt{n}(\bar{\theta} - \theta_0)) + o_P(1) \end{aligned}$$

□

Lemma 2. *Let*

$$(\hat{\eta}, \hat{\lambda}) = \arg \min_{\eta, \lambda} \frac{1}{n} \sum_{i=1}^n f(\eta + \lambda' g(X_i, \bar{\theta}) - \eta).$$

Then, for any $\bar{\theta}$ such that $\bar{\theta} \xrightarrow{P} \theta_0$, such that $\mathbb{E} g(X_i, \theta_0) = 0$, we have that

$$n\hat{\eta} = n\hat{\lambda}'\Omega_0\hat{\lambda} \left(1 - \frac{1}{2}f_3\right) + o_P(1)$$

Proof. The first-order condition for $\hat{\eta}$ is

$$\frac{1}{n} \sum_{i=1}^n f'(\hat{\eta} + \hat{\lambda}' g(X_i, \bar{\theta})) - 1 = 0$$

Since $\bar{\theta} \xrightarrow{P} \theta_0$, we have that by Lemma 1, $\hat{\lambda} \xrightarrow{P} 0$ and $\hat{\eta} \xrightarrow{P} 0$. A second-order expansion in the implicit function $\hat{\eta} := \hat{\eta}(\hat{\lambda})$ about the point $(0, 0)$ gives

$$\hat{\eta}(\hat{\lambda}) = \frac{\partial}{\partial \lambda} \hat{\eta}(0)' \hat{\lambda} + \frac{1}{2} \hat{\lambda}' \frac{\partial^2}{\partial \lambda \partial \lambda'} \hat{\eta}(\bar{\lambda}) \hat{\lambda}$$

Using the implicit function theorem,

$$\frac{\partial}{\partial \lambda} \hat{\eta}(0) = -\frac{1}{f_2} \mathbb{P}_n \bar{g}$$

recalling $f_2 = f''(0) = 1$. Applying the implicit function theorem a second time, we have

$$\frac{\partial^2}{\partial \lambda \partial \lambda'} \hat{\eta}(\bar{\lambda}) = -f_3 \Omega_0 + o_P(1)$$

recalling that $f_3 = f'''(0)$. Combining these expressions, we have

$$\begin{aligned} n\hat{\eta} &= -\mathbb{G}_n \bar{g}' \sqrt{n} \hat{\lambda} - \frac{f_3}{2} n \hat{\lambda}' \Omega_0 \hat{\lambda} + o_P(1) \\ &= n \hat{\lambda}' \Omega_0 \hat{\lambda} \left(1 - \frac{f_3}{2}\right) \end{aligned}$$

□

Lemma 3. *Along the sequence $a(\theta_0) = \delta/\sqrt{n}$, $n\hat{\lambda}'\Omega_0\hat{\lambda} \Rightarrow \chi_r^2(\xi^2(\delta))$.*

Proof. From Newey and McFadden (1994), we have that

$$\sqrt{n}(\bar{\theta} - \theta_0) = -B^{-1}R'(RB^{-1}R')^{-1}\delta - B^{-1/2}MB^{-1/2}G'_0\Omega_0^{-1}\mathbb{G}_n g_0 + o_P(1)$$

Thus, from Lemma 1, we have that

$$\begin{aligned}\sqrt{n}\Omega_0^{1/2}\hat{\lambda} &= -\Omega_0^{-1/2}\mathbb{G}_n g_0 - \Omega_0^{-1/2}G_0\sqrt{n}(\bar{\theta} - \theta_0) + o_P(1) \\ &= -\Omega^{-1/2}\mathbb{G}_n g_0 + \Omega_0^{-1/2}G_0B^{-1}R'(RB^{-1}R')^{-1}\delta \\ &\quad + \Omega_0^{-1/2}G_0B^{-1/2}MB^{-1/2}G'_0\Omega_0^{-1}\mathbb{G}_n g_0 \\ &= \Omega_0^{-1/2}G_0B^{-1}R'(RB^{-1}R')^{-1}\delta \\ &\quad - (I - \Omega_0^{-1/2}G_0B^{-1/2}MB^{-1/2}G'_0\Omega_0^{-1/2})\Omega^{-1/2}\mathbb{G}_n g_0 \\ &\Rightarrow \mathcal{N}\left(\Omega_0^{-1/2}G_0B^{-1}R'(RB^{-1}R')^{-1}\delta, \tilde{M}\tilde{M}\right)\end{aligned}$$

where $\tilde{M} := (I - \Omega_0^{-1/2}G_0B^{-1/2}MB^{-1/2}G'_0\Omega_0^{-1/2})$. Note that $\tilde{M}$ is idempotent:

$$\begin{aligned}\tilde{M}\tilde{M} &= (I - \Omega_0^{-1/2}G_0B^{-1/2}MB^{-1/2}G'_0\Omega_0^{-1/2})(I - \Omega_0^{-1/2}G_0B^{-1/2}MB^{-1/2}G'_0\Omega_0^{-1/2}) \\ &= I - 2\Omega_0^{-1/2}G_0B^{-1/2}MB^{-1/2}G'_0\Omega_0^{-1/2} \\ &\quad + \Omega_0^{-1/2}G_0B^{-1/2}MB^{-1/2}G'_0\Omega_0^{-1}G_0B^{-1/2}MB^{-1/2}G'_0\Omega_0^{-1/2} \\ &= I - \Omega_0^{-1/2}G_0B^{-1/2}MB^{-1/2}G'_0\Omega_0^{-1/2} = \tilde{M}\end{aligned}$$

and thus $\sqrt{n}\Omega^{1/2}\hat{\lambda} \Rightarrow \mathcal{N}(\Omega_0^{-1/2}G_0B^{-1}R'(RB^{-1}R')^{-1}\delta, \tilde{M})$. Furthermore, notice that

$$\begin{aligned}\tilde{M}\Omega_0^{-1/2}G_0B^{-1}R'(RB^{-1}R')^{-1} &= \Omega_0^{-1/2}G_0B^{-1}R'(RB^{-1}R')^{-1} \\ &\quad - \Omega_0^{-1/2}G_0B^{-1/2}MB^{-1/2}G'_0\Omega_0^{-1}G_0B^{-1}R'(RB^{-1}R')^{-1} \\ &= \Omega_0^{-1/2}G_0B^{-1}R'(RB^{-1}R')^{-1} \\ &\quad - \Omega_0^{-1/2}G_0B^{-1/2}MB^{-1/2}R'(RB^{-1}R')^{-1} \\ &= \Omega_0^{-1/2}G_0B^{-1}R'(RB^{-1}R')^{-1} \\ &\quad - \Omega^{-1/2}G_0B^{-1}R'(RB^{-1}R')^{-1} \\ &\quad + \Omega_0^{-1/2}G_0B^{-1}A'_0(A_0B_0^{-1}A_0)^{-1}A_0B_0^{-1}R'(RB^{-1}R')^{-1} \\ &= \Omega_0^{-1/2}G_0B^{-1}A'_0(A_0B_0^{-1}A_0)^{-1}\end{aligned}$$

Furthermore, note that

$$\begin{aligned}
& (\Omega_0^{-1/2} G_0 B^{-1} A'_0 (A_0 B_0^{-1} A_0)^{-1} \delta)' \Omega_0^{-1/2} G_0 B^{-1} A'_0 (A_0 B_0^{-1} A_0)^{-1} \delta \\
& = \delta' (A_0 B_0^{-1} A_0)^{-1} A'_0 B_0^{-1} G'_0 \Omega^{-1} G_0 B_0^{-1} A_0 (A_0 B_0^{-1} A'_0)^{-1} \delta \\
& = \delta' (A_0 B_0^{-1} A_0)^{-1} \delta
\end{aligned}$$

Thus, we can apply Lemma 9.7 in [Newey and McFadden \(1994\)](#), where the generalized inverse of $\tilde{M}$ is I , and obtain that

$$\frac{2}{2 - f_3} n \hat{\eta} = n \hat{\lambda}' \Omega_0 \hat{\lambda} + o_P(1) \Rightarrow \chi_r^2 (\delta' (A_0 B_0^{-1} A_0)^{-1} \delta)$$

□

A.3 Proof of Theorem 3

Now, we consider the case where $a(\theta_0) = \Delta \neq 0$. In this case, $(\hat{\eta}, \hat{\lambda}) \xrightarrow{P} (\eta_0, \lambda_0) \neq (0, 0)$, and $\bar{\theta} \xrightarrow{P} \theta^* \neq \theta_0$, such that $a(\theta^*) = 0$. Considering the behaviour of $\hat{\eta}$: define $c_n = \frac{1}{n} C_\alpha (1 - \frac{1}{2} f'''(0))$. The Type-II error is then

$$P(\hat{\eta} < c_n)$$

Instead of $\hat{\eta}$, let us first consider $\check{\eta}$, defined as

$$\check{\eta} := \arg \min_{\eta} \frac{1}{n} \sum_{i=1}^n f(\eta + \lambda'_0 g(X_i, \theta^*)) - \eta$$

A straightforward application of the Chernoff bound is that for all $t < \eta_0$,

$$\lim_{n \rightarrow \infty} -\frac{1}{n} \log P(\check{\eta} < t) \geq \sup_{\gamma > 0} \{ \gamma - \log \mathbb{E} \exp \{ \gamma (f'(t + \lambda'_0 g(X_i, \theta^*))) \} \} := \Gamma(t)$$

Thus, $\lim_{n \rightarrow \infty} -\frac{1}{n} \log P(\check{\eta} < c_n) \geq \Gamma(0)$.

On the event $|\check{\eta} - \hat{\eta}| < \epsilon$, we have that

$$\lim_{n \rightarrow \infty} -\frac{1}{n} \log P(\hat{\eta} < t, |\check{\eta} - \hat{\eta}| \leq \epsilon) \geq \Gamma(t + \epsilon)$$

Now, consider the case when $|\check{\eta} - \hat{\eta}| > \epsilon$. In particular, we have

$$P(\hat{\eta} < t, |\check{\eta} - \hat{\eta}| > \epsilon) = P(\hat{\eta} < t, \hat{\eta} < \check{\eta} - \epsilon) + P(\hat{\eta} < t, \hat{\eta} > \check{\eta} + \epsilon)$$

The case when $\hat{\eta} > \check{\eta} + \epsilon$ is similar to the case when $\hat{\eta} < \check{\eta} - \epsilon$, so we focus on the latter. Note that our two estimators are solutions to the equations

$$\begin{aligned}\mathbb{P}_n f'(\check{\eta} + \lambda'_0 g_{i,\theta^*}) &= 1 \\ \mathbb{P}_n f'(\hat{\eta} + \hat{\lambda}' g_{i,\bar{\theta}}) &= 1\end{aligned}$$

Since f' is a monotone function, this implies that

$$\begin{aligned}\mathbb{P}_n f'(\hat{\eta} + \epsilon + \lambda'_0 g_{i,\theta^*}) &< 1 \\ \mathbb{P}_n f'(\check{\eta} - \epsilon + \hat{\lambda}' g_{i,\bar{\theta}}) &> 1\end{aligned}$$

Which implies that

$$\mathbb{P}_n f'(\check{\eta} - \epsilon + \hat{\lambda}' g_{i,\bar{\theta}}) - \mathbb{P}_n f'(\hat{\eta} + \epsilon + \lambda'_0 g_{i,\theta^*}) > 0$$

Thus, we have that this implies

$$\begin{aligned}\sup_{\eta_1, \eta_2, \lambda, \theta} |\mathbb{G}_n(f'(\eta_1 - \epsilon + \lambda' g_{i,\theta}) - f'(\eta_2 + \epsilon + \lambda'_0 g_{i,\theta^*}))| \\ > \sqrt{n} \sup_{\eta_1, \eta_2, \lambda, \theta} \mathbb{E} [f'(\eta_2 + \epsilon + \lambda'_0 g_{i,\theta}) - f'(\eta_1 - \epsilon + \lambda' g_{i,\theta^*})] > 0\end{aligned}$$

where the supremum can be taken over $\eta_1, \eta_2 \in \mathbb{R}$, $\lambda \in \mathbb{R}^k$, and $\theta \in \{\theta : a(\theta) = 0\}$. The supremum is then $\|f'\|_\infty$. Observe that since f' is monotone and bounded, $\eta + \lambda'g$ is affine in g , and $\{g(X_i, \theta) : \theta \in \Theta\}$ has a polynomial bracketing or covering number we can use Theorem 2.14.28 in [van der Vaart and Wellner \(2023\)](#), and we have that

$$P(\hat{\eta} - \check{\eta} < -\epsilon) \leq K \exp \{-2n\|f'\|_\infty\}$$

which implies that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log P(\hat{\eta} < t, \hat{\eta} - \check{\eta} < -\epsilon) \leq -2\|f'\|_\infty$$

Comparing the upper bounds we have found, $-\Gamma(t + \epsilon) \leq -2\|f'\|_\infty$, by Lemma 1.2.15 in [Dembo and Zeitouni \(2009\)](#) and continuity of Γ , we then have that

$$\lim_{n \rightarrow \infty} -\frac{1}{n} \log P(\hat{\eta} < c_n) \geq \Gamma(0)$$

B Technical Appendix

Lemma 4. *Suppose that S_n is a statistic such that*

$$S_n = \sum_{i=1}^n Y_{i,n} + R_n$$

where $R_n = o_P(1)$. Now, assume that for $\epsilon > 0$ small enough, there exists a positive sequence a_n and a function $C(\cdot)$, continuous at 0 and increasing, such that

$$\lim_{n \rightarrow \infty} \frac{-\log P(\sum_{i=1}^n Y_{i,n} > d_n - \epsilon)}{a_n} \geq C(-\epsilon).$$

Then,

$$\lim_{n \rightarrow \infty} \frac{-\log P(S_n > d_n)}{a_n} \geq C(0)$$

Likewise, if

$$\lim_{n \rightarrow \infty} \frac{-\log P(\sum_{i=1}^n Y_{i,n} > d_n + \epsilon)}{a_n} \leq C(\epsilon)$$

then

$$\lim_{n \rightarrow \infty} \frac{-\log P(S_n > d_n)}{a_n} \leq C(0).$$

Proof. The proof of the lower bound is similar, so we consider the upper bound. Let $\epsilon > 0$, and suppose that

$$\lim_{n \rightarrow \infty} \frac{-\log P(S_n > d_n)}{a_n} > C(\epsilon).$$

Thus, we can choose a subsequence n_k such that there exists an N such that for all $n_k > N$,

$$\frac{-\log P(S_{n_k} > d_{n_k})}{a_{n_k}} > C(\epsilon). \tag{B.1}$$

Since $R_{n_k} \rightarrow 0$ along this subsequence, there exists a further sub-subsequence such that $R_{n_{k(l)}} \rightarrow 0$ almost surely. Therefore, there exists an N' such that for all $n_{k(l)} > N'$,

$$-\log P(S_{n_{k(l)}} > d_{n_{k(l)}}) \leq -\log P\left(\sum_{i=1}^n Y_{i,n_{k(l)}} > d_{n_{k(l)}} + \epsilon\right)$$

Dividing through by $a_{n_{k(l)}}$ and taking limits, we have:

$$\lim_{n_{k(l)} \rightarrow \infty} \frac{-\log P\left(S_{n_{k(l)}} > d_{n_{k(l)}}\right)}{a_{n_{k(l)}}} \leq \lim_{n_{k(l)} \rightarrow \infty} \frac{-\log P\left(\sum_{i=1}^n Y_{i,n_{k(l)}} > d_{n_{k(l)}} + \epsilon\right)}{a_{n_{k(l)}}} = C(\epsilon).$$

which contradicts (B.1). Therefore, we have that

$$\lim_{n \rightarrow \infty} \frac{-\log P(S_n > d_n)}{a_n} \leq C(\epsilon)$$

and the result follows by continuity of $C(\cdot)$. □